Specialized Scheduler and Notification System

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***Abstract*- Balancing between physiotherapy sessions and school academic load can be challenging or difficult for students living with disabilities especially at King George VI center. To address this issue, we propose a Specialized Scheduler and Notification Mobile Application that allows students and physiotherapist to schedule therapy sessions without conflicting with the academic schedule. The system allows for students to enter their academic schedule, for physiotherapist to view student's free periods and schedule physiotherapy sessions at their free periods, automated reminders of sessions to be send to students and physiotherapist and for students to track their progress of session attendance. The application was built in android studio using Java. The proposed system will provide personalized notifications, scheduling capabilities, and collaboration features. It also supports time management, ensures accessibility, improves organization and planning, enables progress tracking and motivation, and enhances communication and support.**

Keywords: Scheduling, notifications, sessions, progress tracking, communication

I. INTRODUCTION

Students living with physical disabilities at King George VI center often face challenges in managing their healthcare needs (physiotherapy sessions) alongside their academic responsibilities. Physiotherapy sessions are crucial for their well-being, but scheduling and managing these sessions can be difficult due to conflicting academic schedules, forgetfulness, and a lack of personalized time management tools. The proposed tool is a Specialized Scheduler and Notification Mobile Application that allows students and physiotherapist to schedule therapy sessions without conflicting with the academic schedule

II. PROBLEM STATEMENT

Physically challenged students living with disabilities at King Georges VI center often face challenges in managing their time and balancing their school work with physiotherapy sessions. The tend to focus more on their school work load neglecting exercises especially students living with paralysis. They require a system that can provide timely notifications and assist in scheduling their activities effectively. The existing solutions lack customization and fail to cater to the specific needs of these students. Therefore, there is a need for a scheduler and notification application that is specifically designed for these students living with physical disabilities.

III. RELATED WORKS

The SnoopMe Interactive Task Scheduler mobile application[1] by A. Dalvi and I. Siddavatam was designed to assist students in managing their tasks effectively by providing alerts and monitoring their mobile usage. It allows students to schedule tasks, set reminders, and receive notifications about task completion. The app tracks mobile activities and recommends usage restrictions based on the data collected. Students can benefit from graphical representations of their time spent on different apps. Feedback from users indicates positive experiences with the app, with suggestions for additional features to enhance its effectiveness as a student-centric scheduler.

The InvigilatorApp[8], a system designed for efficient scheduling and reminder notifications for invigilation duties in educational institutions. The system automates and optimize the scheduling process, improve communication between administrators and invigilators, and streamline exam administration. Key features include web-based admin and faculty dashboards, real-time notifications, round-robin scheduling, and database management. The system utilizes algorithms such as Particle Swarm Optimization, Constraint Programming, Ant Colony Optimization, Tabu Search, and Hybrid Approaches to enhance scheduling efficiency. The system architecture involves React JavaScript for the user interface and Django framework for the backend, ensuring timely reminders and email notifications.

In [4] an android application designed by D. Ameta, K. Mudaliar, and P. Patel, is focusing on enhancing interactions between doctors and patients while promoting adherence to medication schedules. The application includes a feature that automatically sets off alarms to remind users of their medication timings. Additionally, it allows patients to search for healthcare providers, access health-related information, and receive notifications via email or messages. The system aims to be cost-effective, efficient, and supportive of medication adherence by offering features like facilitating communication between doctors and patients, assisting in medication scheduling, and presenting a user-friendly interface that does not necessitate specialized hardware for operation.

IV. SOLUTION

To solve the problems of the current system of scheduling and notification, A specialized scheduler and notification mobile application is going to be developed with the following objectives:

- To design a user-friendly interface for appointment scheduling that specifically considers the academic schedules of the student to avoid conflicts with physiotherapy sessions.
- To allows the physiotherapists to schedule sessions with their patients/ students
- To allows students to easily communicate with their physiotherapist for clarifications, questions, or updates
- To allow automated reminders through email, or push notifications to ensure students and physiotherapist are reminded of their upcoming physiotherapy sessions.
- To create exercise plans tailored to each student's needs which they can share with their patients
- To allows physically challenged students to securely login and access their personalized physiotherapy exercise plans as shared by their physiotherapist
- To allow users to track and monitor their physiotherapy progress to enable adjustments to their treatment plans as needed.

A. FEATURES OF THE SYSTEM

The proposed system will have the following features:

1. User-friendly interface for appointment scheduling that considers academic schedules to avoid conflicts with physiotherapy sessions.

2. Physiotherapists can schedule sessions with their patients/students.
3. Students can easily communicate with their physiotherapist for clarifications, questions, or updates.
4. Automated reminders through email or push notifications to ensure students and physiotherapists are reminded of their upcoming physiotherapy sessions.
5. Exercise plans tailored to each student's needs which they can share with their patients.
6. Physically challenged students can log in and access their personalized physiotherapy exercise plans as shared by their physiotherapist.
7. Users can track and monitor their physiotherapy progress to enable adjustments to their treatment plans as needed.

B. BENEFITS OF THE SYSTEM

The Proposed system will have the following benefits:

1. The system streamlines appointment scheduling
2. Improves communication between students and physiotherapists
3. Sends automated reminders
4. Allows personalized exercise plans
5. Users can track their physiotherapy progress, allowing physiotherapists to make necessary adjustments based on individual progress, resulting in more effective and personalized care.
6. The system enhances collaboration and attendance among students and physiotherapists.

C. SOLUTION ARCHITECTURE

Presentation Layer- User Interface (UI) for appointment scheduling of the mobile application was developed using Java in android studio

Application Layer – Scheduling Module which manages appointment scheduling, conflicts, and availability, communication modules which facilitates secure messaging between physiotherapists and students. Notification module which sends automated reminders and notifications

Business Logic Layer – Triggers reminders and notifications based on schedules and deadlines

Data Access Layer – Database (MSSQL) which stores user data, schedules, exercise plans and communication records

Infrastructure Layer- Server (on-premises (SQL Server Management Studio Management 19)) hosts the database, notification services (email notifications) to send reminders and notification.

Security – Data Encryption (HTTPS) which protects data in transit and at rest

V. RESULTS AND FUTURE WORKS

A. RESULTS

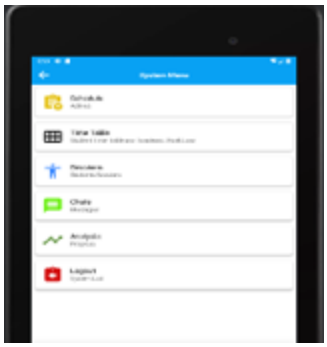

Fig 1. User Interface for users

The system performed well in allowing the physiotherapist to see the student's academic load and schedule for physiotherapy sessions, sending automated reminder to both the student and the physiotherapist, also allowing for communication between the physiotherapist and the student through a messaging platform inbuilt in the platform and the results are summarized in Table I.

Objectives	**Fully Achieved**	**Partially Achieved**
To design a user-friendly interface for appointment scheduling that specifically considers the academic schedules of the student to avoid conflicts with physiotherapy sessions.	✓	
To allows the physiotherapists to schedule sessions with their patients/ students	✓	
To allows students to easily communicate with their physiotherapist for clarifications, questions, or updates	✓	
To allow automated reminders through email, or push notifications to ensure students and physiotherapist are reminded of their upcoming physiotherapy sessions.	✓	
To create exercise plans tailored to each student's needs which they can share with their patients	✓	

To allows physically challenged students to login and access their personalized physiotherapy exercise plans as shared by their physiotherapist	✓	
To allow users to track and monitor their physiotherapy progress to enable adjustments to their treatment plans as needed		✓

B. FUTURE WORKS

Future works of the system include:

1. Integration with Wearable Devices – integrating the app with wearable devices such as fitness trackers or smartwatches to collect real-time data on students' physical activities and progress. This data can be used to further personalize exercise plans and track progress accurately.
2. Incorporate telehealth capabilities into the system, allowing for virtual physiotherapy sessions. This would enable remote consultations and exercises, expanding access to physiotherapy services for students who are unable to attend in-person sessions
3. Enhance the communication module with additional features such as video conferencing, instant messaging, or chatbot support
4. Implementing data analytics capabilities to analyze the collected data and generate insights

VI. CONCLUSION

In conclusion, the proposed Specialized Scheduler and notification application offers a comprehensive solution for managing physiotherapy sessions, communication, and exercise plans for physically challenged students. By considering the students' academic schedule and workload system minimizes conflicts and optimizes session scheduling. The system's user-friendly interface, automated reminders, and secure communication channels enhance the overall experience for both physiotherapists and students. The personalized exercise plans and progress tracking features enable tailored care and informed treatment adjustments. The proposed system architecture ensures

scalability, security, and maintainability, making it a viable solution for educational institutions. By streamlining physiotherapy management, this system has the potential to improve the lives of physically challenged students at King George VI center, enabling them to manage their conditions more effectively and achieve their academic goals. Future development directions include integrating with existing student information systems, expanding to cater to other healthcare professionals, and incorporating artificial intelligence to enhance exercise plan generation and progress tracking. The Scheduler and Notification Application demonstrates the potential of technology in enhancing healthcare management and promoting inclusive education.

ACKNOWLEDGEMENTS

I would like to express my deepest gratitude and appreciation to all those who have supported me throughout the successful completion of this project.

First and foremost, I would like to thank the Lord Almighty, whose presence and grace has been with me from start of this program till now that l am achieving a great milestone.

I would like to thank my supervisor, Ms L Amos for her invaluable guidance, expertise, and continuous support throughout the entire research process. Her insightful feedback, constructive criticism, and encouragement have been instrumental in shaping and refining this project.

I am also immensely grateful to the faculty members of the Software Engineering department, whose teachings and mentorship have provided me with a strong foundation in my field of study. Their passion for academia and dedication to imparting knowledge have been truly inspiring.

I extend my heartfelt thanks to my mother and father for being prayer warriors and providing for me throughout the journey. I am also grateful to my family and friends for their unwavering support and belief in me. Their encouragement, patience, and understanding have been my pillars of strength during the ups and downs of this academic pursuit. I am grateful for their love, encouragement, and motivation that have kept me going.

I would like to acknowledge the participants of my study, whose willingness to share their experiences and insights has contributed significantly to the research findings. Their contributions have been invaluable in shaping the outcomes of this dissertation.

Lastly, I would like to express my profound appreciation to all the individuals who have indirectly contributed to this research through their academic work, publications, and previous studies. Their contributions have been instrumental in shaping the theoretical framework and methodology of this dissertation.

In conclusion, I am deeply thankful to everyone who has played a role, big or small, in the completion of this project. Your support, guidance, and encouragement have been invaluable, and I am truly grateful for the opportunity to have undertaken this research journey with your unwavering support.

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